

COST OF QUALITY (COQ) ANALYSIS REPORT

(Additive Manufacturing Line)

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SECTION 1: HEADER & PROCESS CLASSIFICATION

- Target Benchmark Enterprises: L&T Advanced Manufacturing, Siemens Additive Solutions, & Wipro 3D
- Product Line: 3D Printed Precision Enclosures & Brackets
- Batch Size: 1,000 Units
- Standard Framework: PAF Model (Prevention, Appraisal, Internal Failure, External Failure)

Evaluated AM Categories (ISO/ASTM 52900 Standards):

- Material Extrusion (FDM)
- Powder Bed Fusion (SLS / SLM) — [Primary Selected Process]
- Vat Photopolymerization (SLA)
- Material Jetting (PolyJet)
- Binder Jetting
- Directed Energy Deposition (DED)
- Sheet Lamination (LOM)

Chosen Process: Powder Bed Fusion (Selective Laser Sintering - SLS)

Reason: We selected SLS because it gives strong mechanical parts, needs zero support structure wastage, and gives good dimensional accuracy for industrial enclosures as seen in benchmarks of L&T and Wipro 3D.

SECTION 2: OBJECTIVE & ZERO-DEFECT STRATEGY

Main Objective: Our main goal is to use the PAF model in additive manufacturing to make sure our 3D printing process becomes super effective. We want to build a Zero Defect System so that no failure happens at client side, while saving money for the company at the same time. We are targeting both financial savings and quality improvement together. By spending more money on Prevention and Appraisal early on, we cut down internal scrap and bring external customer failures straight to 0%.

SECTION 3: PAF MODEL COST BREAKDOWN MATRIX

COQ Category	Sl.	Key Quality Activity	Details & Practical Execution Logic	Cost (₹)	%
1. Prevention Costs <i>(Good Quality Cost)</i>	1.1	Workforce Training & Skill Assessments	- First give proper training to all workers and employees on how to operate machines. - Do regular skill assessment sessions to find weak areas. - Make performance and audit reports for each worker to see where they lack and give targeted training to save prevention cost.	₹ 35,000	7.0%
	1.2	CAD Design & FEA Analysis	- Before printing, focus heavily on CAD modeling. - Engineers can analyze stress and strain right inside CAD to fix issues early. - Unlike conventional machining where whole part gets ruined, CAD lets us target and fix the exact spot.	₹ 40,000	8.0%
	1.3	Software Updates & File Formats	- Keep software updated according to new standards. - STL format has issues like not saving colors or textures, so use better formats (like 3MF/AMF) that fulfill all needs without drawbacks.	₹ 25,000	5.0%
	1.4	AI & ML Integration	Integrate AI and Machine Learning into slicers and 3D printing process to make things way more effective and smart.	₹ 30,000	6.0%
	1.5	Environmental & Material Safety	- Check materials for environmental degradation and harmful gas emissions. - Follow air pollution laws so company doesn't face risk or get banned.	₹ 20,000	4.0%
	1.6	Calibration & Root Cause Tools	- Properly calibrate instruments to avoid environmental errors. - Use Root Cause Analysis, Fishbone (Ishikawa), and 5-Why analysis beforehand.	₹ 25,000	5.0%
	TOTAL PREVENTION COST (Money spent to stop errors before printing)			₹ 1,75,000	35.0%
2. Appraisal Costs <i>(Good Quality Cost)</i>	2.1	Raw Material & System Inspection	- Inspect raw material carefully before and after use, as raw material is a major factor. - Inspect the full system, the process happening, and also the people working on it.	₹ 20,000	4.0%
	2.2	FMEA & RPN Allocation	- Find all failure points, accumulate data, and assign Severity, Occurrence, and Detection numbers. - Focus prevention on issues having highest RPN value.	₹ 15,000	3.0%
	2.3	Measurement System Analysis (MSA)	- Apply MSA to inspect instruments and check calibration level. - Use Repeatability and Reproducibility (Gauge R&R)—if readings come close, instrument is good; if not, recalibrate.	₹ 15,000	3.0%
	2.4		Use Control Charts (X-bar and R charts, standard deviation charts) to see if values cross specification	₹ 25,000	5.0%

COQ Category	Sl.	Key Quality Activity	Details & Practical Execution Logic	Cost (₹)	%
		Integrated AI Control Charts	limits. • Use Six Sigma to catch variations. • Build an Integrated Hybrid System where AI checks Six Sigma, FMEA, and Control Charts step-by-step automatically.		
	2.5	Advanced NDT Testing	• Perform NDT testing to check for hidden defects and internal voids. • Use advanced tests like Micro-CT X-Ray scan and Ultrasonic testing.	₹ 35,000	7.0%
	2.6	Dimensional Verification	• Verify first printed article using 3D scanners and CMM to check exact shape and size.	₹ 15,000	3.0%
TOTAL APPRAISAL COST (Money spent on checking, testing, and inspecting)				₹ 1,25,000	25.0%
3. Internal Failure Costs (Bad Quality Cost)	3.1	Warping & Distortion Scrap	• Scrap loss from warping and dimensional distortion occurring during 3D printing.	₹ 50,000	10.0%
	3.2	Stair-stepping & Surface Sanding	• Extra sanding, vapor smoothing, and rework needed due to stair-stepping effect on printed surfaces.	₹ 40,000	8.0%
	3.3	Surface Roughness & Delamination	• Scrapped parts due to rough surface finish and layer bonding failure (especially seen in LOM and extrusion).	₹ 35,000	7.0%
	3.4	Nozzle Clogging & Material Retention	• Problem where material stays inside nozzle causing complete nozzle clogging, print aborts, and lost time.	₹ 40,000	8.0%
	3.5	In-factory Gas Damage	• Fumes and gases coming out during printing that affect and damage factory materials and surrounding tools.	₹ 35,000	7.0%
TOTAL INTERNAL FAILURE COST (Losses and rework inside factory before shipping)				₹ 2,00,000	40.0%
4. External Failure Costs (Bad Quality Cost)	4.1	Customer Rejections & Returns	• Goal achieved: 0% customer rejections because of strong Prevention (35%) and Appraisal (25%).	₹ 0	0.0%
	4.2	Warranty & Brand Claims	• Zero bad parts reached client side. Zero defect system target met.	₹ 0	0.0%
TOTAL EXTERNAL FAILURE COST (Customer level losses - Target Achieved)				₹ 0	0.0%
GRAND TOTAL COST OF QUALITY (COQ) — Full Batch Budget (1,000 Units)				₹ 5,00,000	100.0%

SECTION 4: FINANCIAL & QUALITY IMPACT ANALYSIS

- **Good Quality Investment (Prevention + Appraisal):** ₹ 3,00,000 (60.0%)
- **Bad Quality Cost (Internal Scrap/Rework):** ₹ 2,00,000 (40.0%)
- **External Failure Cost:** ₹ 0 (0.0%)

Summary Analysis: By spending 60% on proper worker training, CAD analysis, AI integration, MSA, Control Charts, and Micro-CT scanning, we stopped defects from reaching the customer. All failure costs stayed inside factory as routine internal scrap (40%), completely eliminating external claims and brand damage.

SECTION 5: RECOMMENDATIONS & FINAL SIGN-OFF

1. **AI Auto-Correction:** Hook up the AI-based hybrid control charts directly with 3D printer controls so it catches warping and nozzle issues early and adjusts settings automatically.
2. **File Format Shift:** Move completely away from basic STL format to formats like 3MF/AMF so texture and color data are not lost during slicing.
3. **Fume Control:** Set up better exhaust and air filtration inside factory so escaping gases don't damage factory equipment or degrade raw materials.

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